Python Course for Physicists (And everyone else!) Chunk 11: Reading and Writing Files

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Focus: Delimiter Separated Values (e.g. csv)

- “Normal” way:
  → Slow
  → No conversion to numpy data

```
with open(filename) as f:
    s = f.read()
```

- Better way, works 90 % of time:
  → Lot of helpful options
  → Generates a numpy array
  → BUT: does not allow “,” as a decimal separator:
  ❌ 3,1415

```
np.loadtxt(filename)
```

- Solution: Pandas

[https://www.spiegel.de/wissenschaft/natur/iucn-ueber-pandabaeren-gefaehrdet-statt-stark-gefaehrdet-a-1110974.html]

```
1880.001,-0.692
1880.004,-0.592
1880.007,-0.673
1880.010,-0.615
1880.012,-0.681
1880.015,-0.743
1880.018,-0.646
1880.021,-0.716
1880.023,-0.984
1880.026,-0.937
1880.029,-1.125
1880.031,-1.283
1880.034,-1.108
1880.037,-0.960
1880.040,-0.865
1880.042,-0.753
1880.045,-0.511
1880.048,-0.678
1880.051,-0.766
1880.053,-0.875
1880.056,-0.913
1880.059,-0.812
1880.062,-0.744
1880.064,-0.762
1880.067,-0.900
1880.070,-1.070
1880.072,-1.260
```

Pandas

- Library for tabular data (built on top of numpy)
  - Can be very useful, but out of scope for this course
  - We only use it for reading data
- Install in Anaconda Powershell Prompt:
  conda install pandas
- To read a table:

```
filename = "data\\daily_temperature_cleaned.txt"
data = pd.read_table(filename, delimiter=',').to_numpy()
print(data)
```

```
[[ 1.880004e+03 -5.920000e-01]
 [ 1.880007e+03 -6.730000e-01]
 [ 1.880010e+03 -6.150000e-01]
 ...
 [ 2.022574e+03  1.574000e+00]
 [ 2.022577e+03  1.577000e+00]
 [ 2.022579e+03  1.629000e+00]]
```

```
1880.001,-0.692
1880.004,-0.592
1880.007,-0.673
1880.010,-0.615
1880.012,-0.681
1880.015,-0.743
1880.018,-0.646
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1880.064,-0.762
1880.067,-0.900
1880.070,-1.070
1880.072,-1.260
```

Relative and Absolute Paths

- Path: String that specifies a file / directory location:
- Absolute Path: Path from “the bottom”:
  `C:\2025 Python Course\Task Sheets\data\oscilloscope.csv`
- Current Working Directory (cwd)
  - cwd: The file path where your code is executed: Usually the parent folder of your executed file:
    `C:\2025 Python Course\Task Sheets\`
  - Find out the cwd:

```
import os
print(os.getcwd())
```

- Relative Path: Path relative to the current working directory:
  `data\oscilloscope.csv`
- File names can be given as a relative or absolute path.
- Careful: “\” is an escape sequence in a string → use “\\” instead.

Live Coding: The Try and Error of File Loading

- Data is often “dirty”:
  - Weird formats
  - Missing values
  - More data than we need
  - Headers and footers
  - Much more stuff, that messes it up
- → Usually, a try and error process:
  → No universal method ☹

```
%
% Results are based on 50461 monthly time series
%   with 21047039 observations and 48263 daily time series
%   with 512331899 observations
%
% Estimated Jan 1951-Dec 1980 land-average temperature (C): 8.59 +/- 0.05
%
%
%
% Date Number, Year, Month, Day, Day of Year, Anomaly

  1880.001     1880     1     1        1      -0.692
  1880.004     1880     1     2        2      -0.592
  1880.007     1880     1     3        3      -0.673
  1880.010     1880     1     4        4      -0.615
  1880.012     1880     1     5        5      -0.681
  1880.015     1880     1     6        6      -0.743
  1880.018     1880     1     7        7      -0.646
  1880.021     1880     1     8        8      -0.716
  1880.023     1880     1     9        9      -0.984
  1880.026     1880     1    10       10      -0.937
  1880.029     1880     1    11       11      -1.125
  1880.031     1880     1    12       12      -1.283
  1880.034     1880     1    13       13      -1.108
```

File Saving

- Human readable: Easy with numpy:

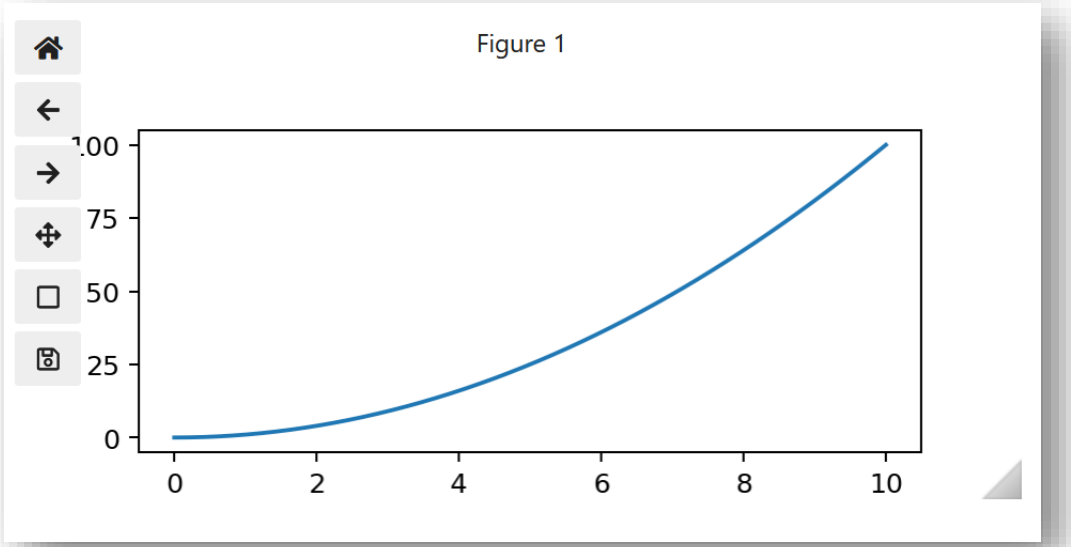
```
data = pd.read_table(load_filename, sep='\\s+', skiprows=22).to_numpy()[:, [0, 5]]
save_filename = "data\\daily_temperature_cleaned.txt"
np.savetxt(save_filename, data, fmt='%.3f', delimiter=",")
```

- Only machine readable:
  → Faster
  → Error proof

```
np.save("data", data)
data = np.load("data.npy")
```

Extra Tip: Interactive Matplotlib

- Matplotlib can be made interactive:
- For this: conda install ipympl



- Before importing: %matplotlib widget:

```
%matplotlib widget
import matplotlib.pyplot as plt
```

Good Luck with your Tasks ☺

Morning: Program works perfectly, no errors

Program in the afternoon:

[https://python.plainenglish.io/20-funny-programming-memes-that-will-make-your-day-5793a8e1608b]